A Few Useful Tricks

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Last update: August 18, 2021

1 Trick 1: $\left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right) = \frac{1}{n} - \frac{1}{n+1}$

This incredibly useful bit of mathematical magic works when the denominators of the two fractions are of the form n and $n+1$, their numerators are both 1, and $n > 0$. For example, something like

$$\left(\frac{1}{3}\right)\left(\frac{1}{4}\right) = \frac{1}{12} \text{ and } \frac{1}{3} - \frac{1}{4} = \frac{4-3}{12} = \frac{1}{12}$$

So why does this work? Consider

$$\begin{aligned}
\left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right) &= \frac{1}{n \cdot (n+1)} && \text{\# numerators are both 1} \\
&= \frac{(n-n)+1}{n \cdot (n+1)} && \text{\# } (n-n) = 0 \text{ and } 0+1 = 1 \\
&= \frac{(n+1)-n}{n \cdot (n+1)} && \text{\# rearrange (+ is commutative and associative)} \\
&= \frac{(n+1)}{n \cdot (n+1)} - \frac{n}{n \cdot (n+1)} && \text{\# split into a difference} \\
&= \frac{1}{n} - \frac{n}{n \cdot (n+1)} && \text{\# cancel } (n+1) \text{ in the first term} \\
&= \frac{1}{n} - \frac{1}{n+1} && \text{\# cancel } n \text{ in the second term } \square
\end{aligned}$$

So we have

$$\left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right) = \frac{1}{n} - \frac{1}{n+1} \tag{1}$$

As an example, let $n = 2$ in Equation 1. Then we get

$$\left(\frac{1}{2}\right)\left(\frac{1}{3}\right) = \frac{1}{6}$$

and

$$\frac{1}{2} - \frac{1}{3} = \frac{3-2}{6} = \frac{1}{6}$$

1.1 What is the Sum of the Reciprocals of the Triangular Numbers?

Equation 1 turns out to be useful when evaluating various types of infinite series such as the telescoping series [3] that we are interested in here. In an infinite telescoping series the partial sums (called telescoping sums) contain pairs of consecutive terms which cancel each other, leaving only the initial and final terms. This cancellation is metaphorically called "collapsing the telescope".

For example, when calculating the infinite sum of Equation 1 (a telescoping series), we see that terms from consecutive indexes, such as $\left(-\frac{1}{2} + \frac{1}{2}\right)$ and $\left(-\frac{1}{3} + \frac{1}{3}\right)$ cancel. This infinite sum can thus be evaluated as follows:

$$\begin{aligned}
\sum_{n=1}^{\infty}\left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right) &= \sum_{n=1}^{\infty}\left(\frac{1}{n} - \frac{1}{n+1}\right) && \text{\# Equation 1 (this is the trick)} \\
&= \lim_{N\to\infty}\sum_{n=1}^{N}\left(\frac{1}{n} - \frac{1}{n+1}\right) && \text{\# take the limit} \\
&= \lim_{N\to\infty}\left[\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{N} - \frac{1}{N+1}\right)\right] \\
&= \lim_{N\to\infty}\left[1 + \left(-\frac{1}{2} + \frac{1}{2}\right) + \left(-\frac{1}{3} + \frac{1}{3}\right) + \cdots + \left(-\frac{1}{N} + \frac{1}{N}\right) - \frac{1}{N+1}\right] \\
&= \lim_{N\to\infty}\left[1 - \frac{1}{N+1}\right] \\
&= 1
\end{aligned}$$

In another example Burkard Polster (aka Mathologer) uses this trick when describing how Euler's solution to the Basel problem works [1].

1.2 Triangular Numbers

The example I want to explore here is the infinite sum of the reciprocals of the triangular numbers [4]. So what is a triangular number? The first six triangular numbers are depicted in Figure 1. We can see from the figure that a triangular number T_n is the sum of the integers from 1 to n.

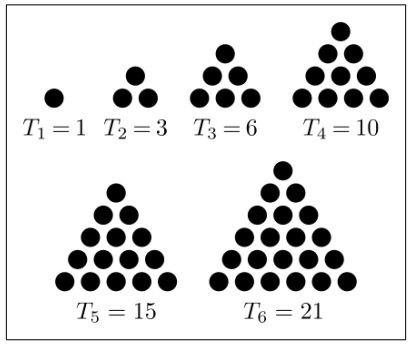

Figure 1: The First Six Triangular Numbers [4]

Since T_n is just the sum of the integers from 1 to n we can easily calculate the first few values of T_n, which are 1, 3, 6, 10, 15, 21, 28, 36, 45, 55, 66, 78, 91, 105 and so on (note that some authors include 0 in the triangular numbers).

Ok, but what is the general form of T_n? Since T_n is the sum of the first n integers we know that

$$T_n = \sum_{k=1}^{n} k = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2} = \binom{n+1}{2} \tag{2}$$

Interestingly if you start with the number 1 in row 2 of Pascal's Triangle [2] the corresponding diagonal contains the triangular numbers. This is shown in blue in Figure 2.

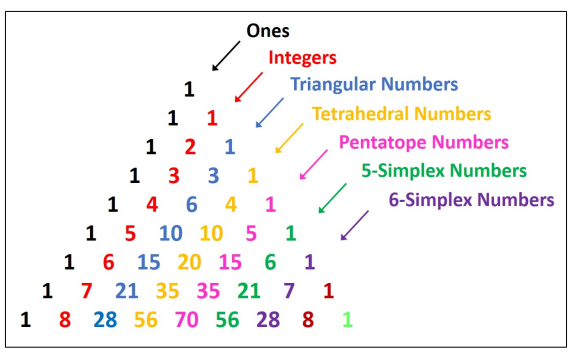

Figure 2: Number Sequences Found In Pascal's Triangle

As an aside, the German mathematician and scientist Carl Friedrich Gauss is said to have discovered the consecutive integer formula, $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$, as a young person; this history is at least controversial [4].

Ok, but what about the sum of the reciprocals of the triangular numbers? This sum looks like

$$\begin{aligned}
\sum_{n=1}^{N} \frac{1}{T_n} &= \sum_{n=1}^{N} \frac{1}{\frac{n(n+1)}{2}} && \text{\# definition of } T_n \text{ (Equation 2)} \\
&= \sum_{n=1}^{N} \frac{2}{n(n+1)} && \text{\# simplify fraction} \\
&= 2\sum_{n=1}^{N} \left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right) && \text{\# pull 2 out of sum, group terms} \\
&= 2\sum_{n=1}^{N} \left(\frac{1}{n} - \frac{1}{n+1}\right) && \text{\# this is the trick (Equation 1)}
\end{aligned}$$

We saw in Section 1.1 that $\lim_{N\to\infty} \sum_{n=1}^{N} \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1$ and we know from the above that the infinite sum of the reciprocals of the triangular numbers equals twice this limit, so we know that

$$\sum_{n=1}^{\infty} \frac{1}{T_n} = 2\left[\lim_{N\to\infty} \sum_{n=1}^{N} \left(\frac{1}{n} - \frac{1}{n+1}\right)\right] = 2$$

1.3 A Slightly More General Form Of This Trick

A slightly more general form of this trick allows for denominators of the form n and $n+k$ rather than the n and $n+1$ denominators we saw in Section 1. The constraints on this form of the trick are that the denominators are of the form n and $n+k$, the product of the numerators must equal the absolute value of the difference of the denominators (that is, k), and $n > 0$.

If we consider the case in which $k = ab$ with $a \geq 0$, $b \geq 0$, and $n > 0$ we see that

$$
\begin{aligned}
\left(\frac{a}{n}\right)\left(\frac{b}{n+ab}\right) &= \frac{ab}{n\cdot(n+ab)} && \text{\# the product of the numerators } ab = |n-(n+ab)| \\
&= \frac{(n-n)+ab}{n\cdot(n+ab)} && \text{\# } (n-n)=0 \text{ and } 0+ab=ab \\
&= \frac{(n+ab)-n}{n\cdot(n+ab)} && \text{\# rearrange (+ is commutative and associative)} \\
&= \frac{(n+ab)}{n\cdot(n+ab)} - \frac{n}{n\cdot(n+ab)} && \text{\# split into a difference} \\
&= \frac{1}{n} - \frac{n}{n\cdot(n+ab)} && \text{\# cancel } (n+ab) \text{ in the first term} \\
&= \frac{1}{n} - \frac{1}{n+ab} && \text{\# cancel } n \text{ in the second term } \square
\end{aligned}
$$

So for $a \geq 0$, $b \geq 0$, and $n > 0$ we have

$$\left(\frac{a}{n}\right)\left(\frac{b}{n+ab}\right) = \frac{1}{n} - \frac{1}{n+ab} \tag{3}$$

To see how this works consider the following example: let $a = 2$, $b = 5$ and $n = 3$. Then

$$\left(\frac{2}{3}\right)\left(\frac{5}{3+(2\cdot 5)}\right) = \left(\frac{2}{3}\right)\left(\frac{5}{3+10}\right) = \left(\frac{2}{3}\right)\left(\frac{5}{13}\right) = \frac{10}{39}$$

and

$$\frac{1}{3} - \frac{1}{3+(2\cdot 5)} = \frac{1}{3} - \frac{1}{3+10} = \frac{1}{3} - \frac{1}{13} = \frac{13}{39} - \frac{3}{39} = \frac{13-3}{39} = \frac{10}{39}$$

We can also note that if $a = b = 1$ in Equation 3 we get the special case we considered in Section 1:

$$\left(\frac{1}{n}\right)\left(\frac{1}{n+(1\cdot 1)}\right) = \left(\frac{1}{n}\right)\left(\frac{1}{n+1}\right) = \frac{1}{n} - \frac{1}{n+(1\cdot 1)} = \frac{1}{n} - \frac{1}{n+1}$$

Equation 3 also works correctly for $a = 0$ and $b \neq 0$, $a \neq 0$ and $b = 0$, and $a = b = 0$.

Is this generalization useful? I don't know but it was fun trying to make it work.

2 Conclusions

I'm not sure if Equation 3 is useful for anything but it was a fun exercise to do. On the other hand, the basic trick (Equation 1) turns out to be very useful, especially when evaluating infinite telescoping series.

References

[1] Burkard Polster (aka Mathologer). Euler's real identity NOT e to the i pi = -1. https://www.youtube.com/watch?v=yPl64xi_ZZA, Aug 2017. [Online; accessed 17-August-2021].

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[3] Wikipedia contributors. Telescoping Series — Wikipedia, The Free Encyclopedia. https://en.wikipedia.org/w/index.php?title=Telescoping_series&oldid=1028468162, 2021. [Online; accessed 17-August-2021].

[4] Wikipedia contributors. Triangular Number — Wikipedia, The Free Encyclopedia. https://en.wikipedia.org/w/index.php?title=Triangular_number&oldid=1034664425, 2021. [Online; accessed 17-August-2021].